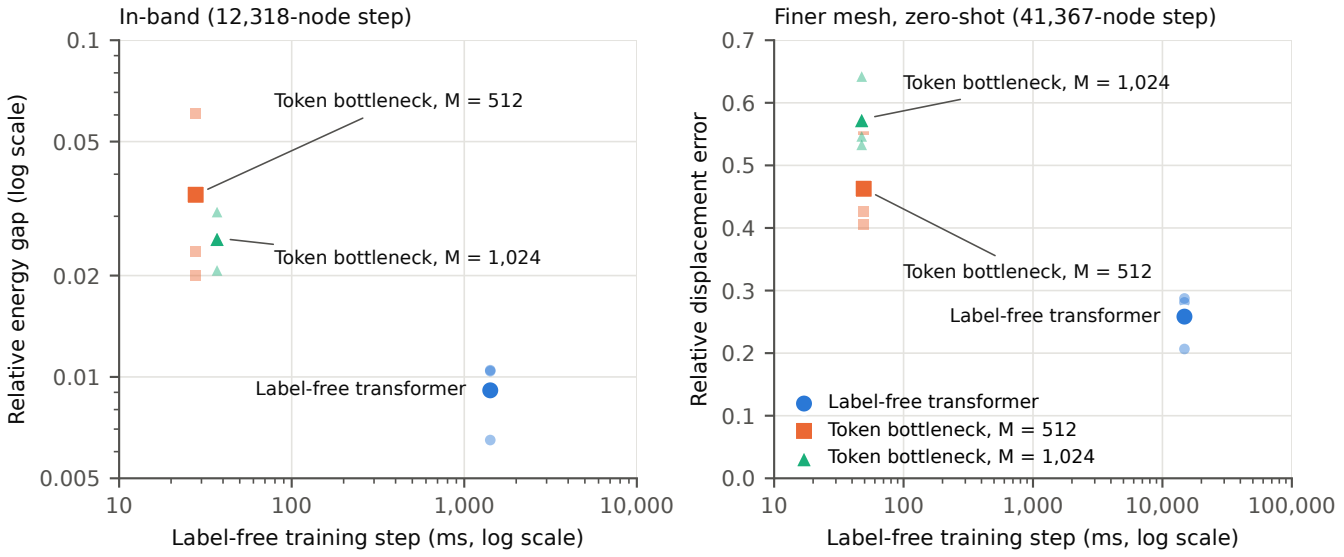
In-band (12,318-node step)
Relative energy gap (log scale)
0.1
0.05
0.02
0.01
0.005
Token bottleneck, M = 512
Token bottleneck, M = 1,024
Label-free transformer
10
100
1,000
10,000
Label-free training step (ms, log scale)
Finer mesh, zero-shot (41,367-node step)
Relative displacement error
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Token bottleneck, M = 1,024
Token bottleneck, M = 512
Label-free transformer
Label-free transformer
Token bottleneck, M = 512
Token bottleneck, M = 1,024
10
100
1,000
10,000
100,000
Label-free training step (ms, log scale)